

JEAN CLAUDE MANIRIHO

Master’s in Mathematical Sciences (Epidemiology) | AIMS Rwanda | Climate & Public Health Researcher | Full-Stack Developer

 jeanclaude.maniriho@aims.ac.rw

 jean-claude-maniriho-3a308929b

 Claude-2Rachande.github.io

 +250 781 556 924

 Claude-2Rachande

 Kigali, Rwanda



PROFESSIONAL SUMMARY

Mathematical Epidemiologist and Data Scientist specializing in statistical modeling, machine learning, AI, and public health research. Experienced in climate-health analytics, epidemiological modeling, and full-stack software development, applying data-driven solutions to address health and environmental challenges. Passionate about leveraging data, technology, and research to support evidence-based decision-making and sustainable impact.

EDUCATION

M.Sc. in Mathematical Sciences (Epidemiology)

AIMS Rwanda

 2025 – Present  Kigali, Rwanda

Mastercard Foundation Scholar | All coursework completed; currently finalizing thesis research

B.Sc. (Hons) Mathematics and Statistics

University of Rwanda, College of Sciences and Technology

 Completed 2025  Kigali, Rwanda

- Specialization: Mathematics-Statistics
- Final Year Project: Survival Analysis Models for Maternal Mortality Risk Assessment in Rwanda
- Conducted survival analysis in R to investigate factors associated with maternal mortality and healthcare outcomes

EXPERIENCE

Research Assistant (Intern)

Kenya Medical Research Institute (KEMRI)

 2026 – Present  Nairobi, Kenya

- Support climate and health research on malaria risk assessment; manage, clean, and analyze health, environmental, and climate datasets across 47 Kenyan counties (62,000+ records).
- Assist in spatial and epidemiological analyses; prepare reports and research documentation.

Founder & Lead – Initiative in Establishment

Safe Childhood, Educated Futures Initiative (SCEFI)

 2025 – Present  Kigali, Rwanda

- Founded a non-profit development initiative dedicated to child education, protection, and community development in under-served Rwandan communities.
- Developed grant proposals and resource mobilisation briefs targeted at international donors and development partners.
- Designed a trilingual website (English, Kinyarwanda, Kiswahili) with Stripe and MTN Mobile Money payment integration.

LIFE PHILOSOPHY

A mind *sharpened*
reshapes **life** and
world alike.

TECHNICAL SKILLS

Python

R

SQL

Java

HTML

CSS

Django

Power BI

SPSS

MATLAB

GitHub

Git Bash

Linux

LaTeX

Machine Learning

Data Visualization

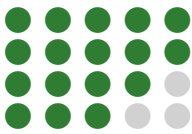
R-INLA

GAMM

Spatial Analysis

LANGUAGES

Kinyarwanda
English
Kiswahili
French



CERTIFICATIONS

AI, Machine Learning and Big Data Analytics

African Engineering and Technology (AFRETEC)

Python Programming
CODTECH IT Solutions

Data Science with R Programming
Simplilearn SkillUp

Data Visualization with Python
IBM Cognitive Class

Climate and Risks Management

- Developed a District Need Index (DNI) methodology to prioritise underserved communities.

Data Analytics Intern
CX Touchpoints Group

- 📅 Mar 2025 – Jun 2025 📍 Kigali, Rwanda
- Collected, cleaned, and analyzed customer experience data to identify trends and service improvements.
 - Developed dashboards and reports using R and Power BI.

Data Analyst Intern
Rwanda Stock Exchange

- 📅 Dec 2024 – Feb 2025 📍 Kigali, Rwanda
- Analyzed stock market and financial datasets; produced analytical reports and supported real-time market monitoring.

Community Mentor (Volunteer)
Rural Education Support Initiative

- 📅 2023 – Present 📍 Rwanda
- Mentored students in Mathematics and Physics, promoting STEAM education and exam preparation.

PROJECTS

Climate Change and Malaria Risk Assessment among School Children in Kenya
KEMRI & AIMS Rwanda

- 📅 2026 – Present 📍 Kenya
- Supporting analysis to assess effects of climate change indicators on the spatial distribution of malaria infection among school children in Kenya.
 - Applying statistical and spatial analysis techniques (R, Python) to identify malaria risk patterns and inform public health interventions.

Modelling *Haemophilus influenzae* Presence in Children with Otitis Media using GLMM
AIMS Rwanda

- 📅 2026 📍 Rwanda
- Analyzed longitudinal clinical trial data (50 children, 220 repeated binary measurements) to evaluate antibiotic efficacy against *Haemophilus influenzae* in pediatric otitis media.
 - Applied a Generalized Linear Mixed Model (Bernoulli family, logit link, random intercept per subject) in R to account for within-subject correlation.

AIMS Workspace Management System
Academic Database Management Project

- 📅 2025 📍 AIMS Rwanda
- Designed a student information management system (Django web app) with a relational database for records, cohorts, programs, and enrollments.
 - Applied SQL, database design, and data modeling to ensure data integrity and efficient retrieval.

United Nations Institute for Training and Research (UNITAR)

LEADERSHIP

Founder & Lead
Safe Childhood, Educated Futures Initiative (SCEFI)

Class Representative
Mathematics-Statistics, University of Rwanda

Activities & Events Coordinator
Mastercard Foundation Scholars Program, AIMS Rwanda

REFERENCES

Prof. Blaise Tchapnda
Academic Director, AIMS Rwanda

✉ blaise.tchapnda@aims.ac.rw
☎ +250 781 446 600

Dr. Marie Yvonne Uwera
Academic Coordinator, AIMS Rwanda

✉ marie.uwera@aims.ac.rw
☎ +250 788 871 566

Dr. Arsénio Nhacolo
Head of Data Management and Statistical Analysis, Manhica Health Research Centre

✉ arsenio.nhacolo@manhica.net

Prof. Ngaruye Innocent
Lecturer, University of Rwanda

✉ ingaruye@gmail.com
☎ +250 788 501 160